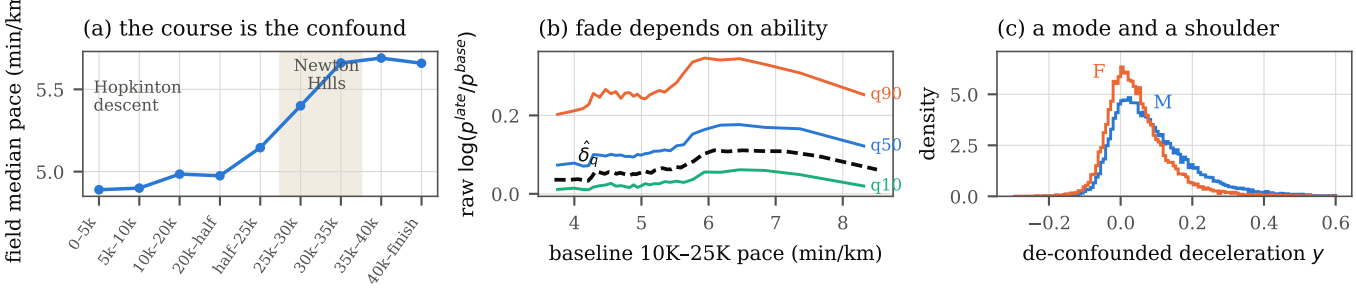
(a) the course is the confound
field median pace (min/km)
5.5
5.0
Hopkinton
descent
Newton
Hills
0–5k
5k–10k
10k–20k
20k–half
half–25k
25k–30k
30k–35k
35k–40k
40k–finish
(b) fade depends on ability
raw log(p^late/p^base)
0.2
0.0
q90
q50
q10
δ̂_q
4
5
6
7
8
baseline 10K–25K pace (min/km)
(c) a mode and a shoulder
density
5.0
2.5
0.0
F
M
−0.2
0.0
0.2
0.4
0.6
de-confounded deceleration y